

**2.13**

Exponential and Logarithmic Equations and Inequalities

Core-to-Challenge Worksheet and AP Practice

AP Precalculus - Unit 2 | ECHS Mathematics | Teacher: Mohammad Abu-Ghuwaleh

Student Information

Name: _____ Class: _____ Date: _____

Directions

Show mathematical evidence. Retain domain restrictions, label graphs, include units, and write complete interpretations. Calculator use is permitted only when the item explicitly requires approximation, regression, or numerical solving.

Readiness

1. Write one precise sentence explaining this principle: Use common bases when possible; otherwise apply logarithms.

2. Write one precise sentence explaining this principle: For logarithmic equations, state domain restrictions before solving.

3. Write one precise sentence explaining this principle: An increasing function preserves inequality direction; a decreasing function reverses it.

4. Write one precise sentence explaining this principle: Every proposed logarithmic solution must make all original arguments positive.

Core

5. Solve $3^{2x-1} = 27$.

6. Solve $2^x = 7$.

7. Solve $\ln(x-1) + \ln(x+1) = \ln 8$.

8. State what the formula $b^u = b^v \Rightarrow u = v$ tells you and name any required restriction.

9. State what the formula $b^x = A \Rightarrow x = \frac{\ln A}{\ln b}$ tells you and name any required restriction.

10. State what the formula $\log_b u = \log_b v \Rightarrow u = v$ ($u, v > 0$) tells you and name any required restriction.

Medium

11. Solve $2^{x+1} = 16$, checking restrictions.

12. Solve $5^{2x} = 7$, checking restrictions.

13. Solve $\log_3(x-1) = 2$, checking restrictions.

14. Solve $\ln x + \ln(x-2) = \ln 3$, checking restrictions.

15. Solve $4^x - 5 \cdot 2^x + 4 = 0$, checking restrictions.

Hard

16. Solve $3^x < 20$ exactly as an interval.

17. Solve $(1/2)^{x-1} \geq 8$.

18. Solve $\log_2(x + 3) > 1$.

19. Solve $\ln(x - 1) = \ln(7 - x)$.

Difficult

20. Solve $2^x = x + 1$.

21. Solve $\log_2 x + \log_2(x - 2) = 3$.

22. Solve $e^{2x} - 6e^x + 5 \leq 0$.

Challenge / HOT

23. Prove $a^x = b$ has a unique real solution for $a > 0, a \neq 1, b > 0$.

24. Find all real x satisfying $\ln(x^2 - 1) = \ln(3x + 1)$.

AP-Style Multiple Choice

25. Solve $3^x = 27$

- (A) 2
- (B) 3
- (C) 9

- (D) 27
26. When bases match and are valid, equate
- (A) coefficients
 - (B) exponents
 - (C) domains only
 - (D) logs to zero
27. Before solving log equation, state
- (A) degree
 - (B) argument domain restrictions
 - (C) vertex
 - (D) ratio
28. For base $0 < b < 1$, inequalities
- (A) preserve exponent direction
 - (B) reverse exponent direction
 - (C) have no solution
 - (D) become quadratic
29. Solve $\ln x = 2$
- (A) $x = 2e$
 - (B) $x = e^2$
 - (C) $x = \ln 2$
 - (D) $x = 2$
30. Extraneous log solutions arise because
- (A) logs are linear
 - (B) algebra may produce nonpositive arguments
 - (C) base is positive
 - (D) x is integer

AP-Style Free Response

31. Solve and justify: (a) $9^x - 10 \cdot 3^x + 9 = 0$; (b) $\log_3(x+1) + \log_3(x-3) = 2$; (c) $(1/4)^x > 8$.

32. A model $P(t) = 500(1.08)^t$. (a) Solve when P exceeds 1000. (b) Find first integer year. (c) Solve when P is between 700 and 900.

Reflection

Identify one item where a domain restriction, unit, assumption, or representation changed your reasoning. Explain in 2-3 sentences.

Original ECHS questions. Selected concepts are informed by Pearson and Holt Precalculus; no textbook exercises are reproduced.